

ZnO nanodispersion as pseudohomogeneous catalyst for alcoholysis of polyethylene terephthalate

Jin-Tao Du^{a,b}, Qian Sun^a, Xiao-Fei Zeng^b, Dan Wang^{a,b}, Jie-Xin Wang^{a,b,*}, Jian-Feng Chen^{a,b}

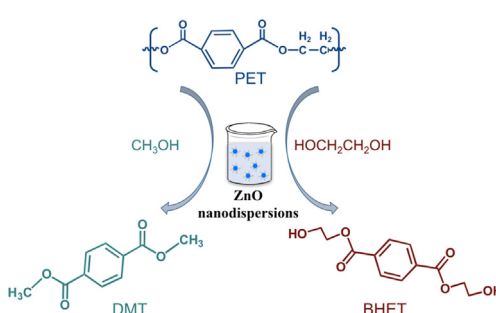
^a Beijing Advanced Innovation Center for Soft Matter Science and Engineering, State Key Laboratory of Organic-Inorganic Composites, Beijing University of Chemical Technology, Beijing 100029, PR China

^b Research Center of the Ministry of Education for High Gravity Engineering and Technology, Beijing University of Chemical Technology, Beijing 100029, PR China

HIGHLIGHTS

- The highly stable dispersions of ultrasmall ZnO nanocrystals were prepared.
- They were used as pseudohomogeneous catalyst for chemical depolymerization of PET.
- ZnO nanodispersions displayed an excellent catalytic activity at 170 °C.
- The methanolysis had shorter reaction time and higher activity than the glycolysis.

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:

Received 20 January 2020

Received in revised form 10 March 2020

Accepted 18 March 2020

Available online 19 March 2020

Keywords:

Pseudohomogeneous catalysts

ZnO nanodispersions

Ultrasmall nanoparticles

Depolymerization of PET

Methanolysis

Kinetics

ABSTRACT

Chemical depolymerization and recycling of polyethylene terephthalate (PET) is a sustainable way to preserve the resources and protect the environment. In this work, methanol and ethylene glycol dispersions of ultrasmall ZnO nanoparticles are firstly adopted as pseudohomogeneous catalysts for alcoholysis of PET. The as-prepared ZnO nanoparticles have a uniform size of 4 nm and can be stable in dispersions for 6 months. In the methanolysis process of PET, the effects of various parameters on the conversion of PET and the yield of dimethyl terephthalate (DMT) were investigated. The results show that higher temperature (170 °C) was beneficial to the conversion of PET and the yield of DMT, which can reach about 97% and 95% after 15 min, respectively. The excellent activity of 553 g PET h⁻¹ (g ZnO)⁻¹ was achieved. Furthermore, the methanolysis of PET have shorter reaction time (1/4) and higher activity (4.7 times) than the glycolysis of PET.

© 2020 Elsevier Ltd. All rights reserved.

1. Introduction

The tremendous increase in the consumption of polyethylene terephthalate (PET) has caused serious environmental problems (Al-Sabagh et al., 2016a; Awaja and Pavel, 2005; Du et al., 2016; Joo et al., 2018; Monsigny et al., 2018; Zhang et al., 2018), and

chemical depolymerization of PET into useful feedstock is an important approach to green sustainability (George and Kurian, 2014; Karayannidis and Achilias, 2007; Paszun and Szychaj, 1997), which can convert plastics into their corresponding monomers or other valuable products (Iannone et al., 2017; Polk et al., 1999). The ester bonds of polymers are cleaved and superseded with the help of a solvent (Geyer et al., 2016; Lorenzetti et al., 2006). Based on different solvents, the chemical depolymerization of PET is divided as methanolysis (Malik et al., 2016), glycolysis (Al-Sabagh et al., 2015; López-Fonseca et al., 2010; Veregue et al., 2018; Wang et al., 2015), hydrolysis (Hoang et al., 2018; Wan

* Corresponding author at: Beijing Advanced Innovation Center for Soft Matter Science and Engineering, State Key Laboratory of Organic-Inorganic Composites, Beijing University of Chemical Technology, Beijing 100029, PR China.

E-mail address: wangjx@mail.buct.edu.cn (J.-X. Wang).

et al., 2001; Wang et al., 2019) and aminolysis (Yamaye et al., 2002). Among them, the methanolysis of PET, which includes the catalytic methanolysis and supercritical methanolysis, has attracted tremendous attention owing to its convenience in operation and purification (Kurokawa et al., 2003). However, the methanolysis of PET under supercritical conditions is generally performed at high-temperature ($>250\text{ }^{\circ}\text{C}$) and high-pressure condition despite the high conversion of PET (Genta et al., 2010; Goto et al., 2002). In this respect, the low temperature catalytic methanolysis of PET is one of the most promising strategies to solve the above-mentioned issues. The catalysts can interact with the PET carbonyl's oxygen to promote the penetration of methanol, which is conducive to reducing the temperature and saving energy.

Presently, the commonly-used transesterification catalysts for the depolymerization of PET mainly include homogeneous catalysts of metal acetates (Zn, Pb, Mn, Co, etc) (Kurokawa et al., 2003; Mishra and Goje, 2003; Yamaye et al., 2002) and heterogeneous catalysts of metal oxides (ZnO , Mn_3O_4 , MgO , TiO_2 , Fe_2O_3 , etc) (Genta et al., 2010; Goto et al., 2002; B. Lin et al., 2019a; Z. Lin et al., 2019b), which have been widely applied to degrade organic pollutants (Ali et al., 2017). Among them, zinc-based catalysts exhibit higher activity towards the depolymerization of PET (Genta et al., 2010; Kurokawa et al., 2003; Yamaye et al., 2002). However, for homogeneous catalysts, their residuals left in the final products may have a great effect on the quality of the products, and the process for separating the catalyst from the products is intricate (Chen et al., 2012; Genta et al., 2005; Yang et al., 2002). For solid heterogeneous catalysts, their poor dispersivity and large size will limit the complete contact and hinder the mass transfer between PET and catalysts, which is considered as the main weakness for using these catalysts. Therefore, it is necessary and urgent to develop a new kind of catalyst, which is easily recoverable, low-temperature and efficient to resolve the above-mentioned problems. Pseudohomogeneous catalysts refer to the uniform dispersion of metal nanoparticles in solvents and generally protected by a stabilizer. If metal active component such as ZnO are made into ultrasmall nanoparticles, which are well dispersed and stabilized in the solvents used in chemical depolymerization, pseudohomogeneous nanocatalysis can be well achieved, thereby completely exhibiting their excellent catalytic activity for depolymerization of PET.

To our best knowledge, there are no reports of the utilization of pseudohomogeneous nanocatalysts in the depolymerization of PET. Previous studies have shown the formation of a porous structure at the external area of PET during the depolymerization process (Chen et al., 2012; Chen et al., 2015). It can be envisioned that pseudohomogeneous nanocatalysts with ultrasmall size can easily enter the porous framework of PET to accelerate the reaction, which is difficult for other heterogeneous catalysts with a larger size and severe agglomeration. Compared with homogeneous catalysts, the heterogeneity of pseudohomogeneous nanocatalysts increases the separability, which combines the advantages of heterogeneous catalysis and homogeneous catalysis. Hence, pseudohomogeneous catalysts are highly recommended to enrich the applicability of the process.

Herein, methanol and ethylene glycol (EG) dispersions of ZnO nanoparticles are firstly adopted as pseudohomogeneous catalysts for alcoholysis of PET. In this catalytic methanolysis process, the effects of reaction parameters including reaction temperature, reaction time, concentration of catalyst and mass ratio of methanol to PET on the conversion of PET, the selectivity of bis(hydroxyethyl terephthalate) (BHET), 2-hydroxyethyl methyl terephthalate (MHET), and the yield of dimethyl terephthalate (DMT) were studied. Furthermore, the possible catalytic mechanism was also explored.

2. Experimental section

2.1. Chemicals

Potassium hydroxide (KOH), zinc acetate dihydrate ($\text{Zn}(\text{Ac})_2 \cdot 2\text{H}_2\text{O}$), methanol, ethanol, trichloromethane (CHCl_3), N-methyl pyrrolidone (NMP) and ethylene glycol (EG) were obtained from Beijing Tong Guang Fine Chemical Co., Ltd. (China). γ -Methacryloxypropyltrimethoxysilane (KH570) and 3-Aminopropyltriethoxysilane (KH550) were provided by Alfa Aesar Chemical Co., Ltd. (China). PET powders ($d \approx 125\text{ }\mu\text{m}$) were supplied by Dongguan Guanju Plastic Co., Ltd. All the reagents were analytically pure and used as received. Deionized water was obtained by a water purification system (RO-DI plus, Hitech, PRC) and used throughout the experiments.

2.2. Preparation of the dispersions of ZnO nanoparticles (ZnO nanodispersions)

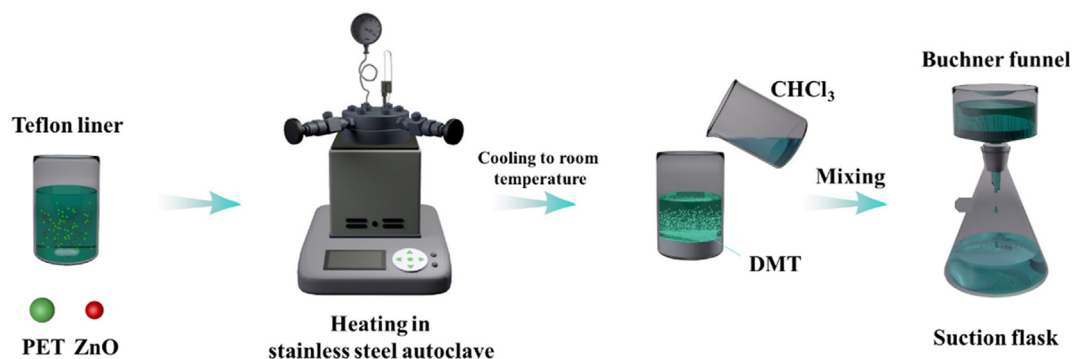
In a typical synthesis of methanol dispersion of ZnO nanoparticles, 42.5 mL $\text{Zn}(\text{Ac})_2 \cdot 2\text{H}_2\text{O}$ methanol solution (0.44 mol/L) was pre-heated in air under mechanical stirring (400 rpm) at $70\text{ }^{\circ}\text{C}$ for 10 min. Subsequently, 42.5 mL KOH methanol solution (0.78 mol/L) was added to the above solution. After 3 min, 20 mL methanol solution containing KH570 (1.4 g) was further added dropwise to the obtained mixture. After the addition of modifier, the reaction was allowed to proceed at $70\text{ }^{\circ}\text{C}$ for 30 min and then ended by the addition of 8 mL deionized water. Stirring and heating are also stopped. After being placed overnight, 50 mL deionized water was added into the solution. The product was collected via filtration and further washed by the mixture of deionized water and ethanol (1:1, v/v) three times. The obtained filter cake was dispersed in methanol, followed by rotary evaporation under $45\text{ }^{\circ}\text{C}$ to remove water and ethanol. Finally, the product was dispersed in methanol again, thereby obtaining transparent methanol dispersion of ZnO nanoparticles.

EG dispersion of ZnO nanoparticles was similarly achieved by replacing KH570 with the mixture of KH570 (0.4 g) and KH550 (1.0 g). After being placed overnight, the product was collected via filtration and further washed by ethanol for three times. The obtained filter cake was dispersed in 3:1 (v/v) water/ethanol mixed solution. After adding EG to the above solution, rotary evaporation was conducted to remove water and ethanol under $60\text{ }^{\circ}\text{C}$ to obtain transparent EG dispersion of ZnO nanoparticles.

As a comparison, untreated ZnO nanoparticles were synthesized under the same conditions except the addition of modifiers. After being placed overnight, untreated ZnO nanoparticles were collected via filtration and further washed by ethanol for three times. The product was dried in a vacuum oven ($40\text{ }^{\circ}\text{C}$) and used for the next step.

2.3. Catalytic activity studies

The experimental process for the methanolysis of PET is schematically shown in Scheme 1. In a typical methanolysis process, 10 g PET, 60 g methanol (75.8 mL) and 0.35 g ZnO nanodispersions (solid content of 20 wt%) were loaded into a Teflon-lined, stainless steel autoclave and heat treated ($120\text{--}170\text{ }^{\circ}\text{C}$) for a time under magnetic stirring (500 rpm). After the reaction was finished, the autoclave was naturally cooled to room temperature, and the DMT monomer would be precipitated during the cooling process. The products were dissolved by the addition of trichloromethane. Finally, the obtained mixture was filtered, and the filtrate was analyzed by a high performance liquid chromatography (HPLC). The



Scheme 1. The schematic diagram of the experimental process for the methanolysis of PET.

selectivity of BHET, MHET and DMT is quantified by peak area normalization method. The yield of DMT is calculated using the following equation:

$$\text{yield of DMT(y\%)} = \frac{\text{conversion of PET(\%)} \times \text{selectivity of DMT(\%)}}{100} \quad (1)$$

Furthermore, the residue was washed with ethanol, dried and weighed as a recovered PET. The depolymerization of PET was calculated using the following equations:

$$\text{Conversion of PET} = \left(1 - \frac{\text{wt of recovered PET}}{\text{wt of initial PET}}\right) \times 100 \quad (2)$$

In a typical glycolysis process, methanol dispersion of ZnO nanoparticles and methanol were replaced with EG dispersion of ZnO nanoparticles and EG, respectively. After the reaction was finished, NMP was applied to dissolve the precipitated BHET with other conditions unchanged. The processing method of the filtrate and residue was the same as the strategy described above.

Furthermore, the recyclability of pseudohomogeneous ZnO nanocatalysts was performed. 10 g PET, 60 g (75.8 mL) methanol and 0.35 g ZnO nanodispersions with a solid content of 20 wt% were reacted at 170 °C for 15 min. After the reaction was completed, the precipitated DMT and unreacted PET were filtrated through a 15-μm membrane. Due to the loss in the reaction and transfer, the methanol solution, in which ZnO is dispersed, was added with more methanol to get 75.8 mL volume. This methanol solution was reused to catalyze the depolymerization of PET in the next cycle. The filtered residue, which contains the precipitated DMT and unreacted PET, was added into 100 mL trichloromethane to dissolve DMT. Unreacted PET was filtrated, washed, dried and weighed to get the conversion of PET. The degradation reaction was carried out for five times in total.

2.4. Characterization

Transmission electron microscope (TEM), high resolution TEM (HRTEM) and selected area electron diffraction (SAED) images of ZnO nanoparticles were obtained using HITACHI, H-9500 transmission electron microscope operating at an accelerating voltage of 300 kV. X-ray diffraction (XRD) patterns of ZnO nanoparticles were carried out by Bruker diffractometer working at 40 kV and 40 mA equipped with CuKα radiation source. The angle (2θ) was measured between 10° and 80° with a scanning rate of 5°/min. The Fourier transform infrared (FTIR) analysis was recorded in a Nicolet 6700 spectrometer (Nicolet Instrument Co., USA) using KBr pellets as a reference material. The X-ray photoelectron spectroscopy (XPS) analysis was conducted on AXIS Supra (Shimadzu, Japan) spectrometer equipped with an Al Kα X-ray excitation (600 W) and

using amorphous C 1s signal at 284.6 eV. Scanning electron microscopy (SEM) images were taken by using a HITACHI, S4800 operating at an acceleration voltage of 20 kV. Thermo gravimetric (TG), derivative thermogravimetry (DTG) analysis and differential scanning calorimeter (DSC) were carried out on TGA/DSC 1/1600 (METTLER TOLEDO) with the heating rate of 10 °C/min from 30 to 700 °C under nitrogen atmosphere. High performance liquid chromatography (HPLC) analysis of methanolysis products was obtained using a Waters 2695–2489 high performance liquid chromatography. A reverse-phase C18 column was used and an ultraviolet (UV) detector was set at 254 nm. A 70:30 (v/v) methanol/H₂O mixed solution was used as the mobile phase at a flow rate of 0.5 mL/min. ¹H and ¹³C nuclear magnetic resonance (NMR) spectra were performed by a Bruker Avance 400 spectrometer. DMT powders were dissolved in chloroform-d (CDCl₃) which contains 0.03% tetramethyl silane (Shanghai Titan scientific Co., Ltd.).

3. Results and discussion

Fig. 1 shows the representative TEM images, HRTEM images, SAED image, digital photographs, corresponding particle size distributions, XRD patterns and FTIR spectra of methanol and EG dispersion of ZnO nanoparticles. It can be obviously seen that the as-prepared ZnO nanoparticles are nearly monodispersed and have an average particle size of about 3.8 nm with a narrow size distribution (**Fig. 1a–c**). Under the influence of surface modifiers (KH570), the methanol dispersion of ZnO nanoparticles has good transparency and could be stable for over 6 months (**Fig. S1**). The ZnO nanoparticles dispersed in EG are similar to the above ZnO nanoparticles dispersed in methanol (**Fig. 1d–f**). In addition, XRD patterns of ZnO nanoparticles show that all the peaks can be well assigned to the hexagonal phase ZnO (JCPDS 36–1451) (**Fig. 1g**), which is in a good agreement with the SAED data. The main diffraction peaks at around 32°, 34°, 36°, 47°, 57°, 63° and 66° were assigned to (1 0 0), (0 0 2), (1 0 1), (1 0 2), (1 1 0), (1 0 3) and (1 1 2) planes. Moreover, the diffraction peaks become considerably broadened, implying a decrease in particle sizes. The FTIR spectra of surface treated ZnO nanoparticles (**Fig. 1h**) show that characteristic peaks at around 1720–1711, 1641 and 1023 cm^{−1} (curve a and b), corresponding to C=O, C=C and Si–O–Si stretching bonds, could be ascribed to the successful modification of KH570 and KH550 on the surfaces of ZnO nanoparticles. The XPS spectra of ZnO nanoparticles also prove the existence of silicon from the silane coupling agent (**Fig. S2**). The above results indicate that the change of surface modifier from KH570 to the mixture of KH570 and KH550 has little effect on the morphology, size and crystal form of ZnO nanoparticles. The amino (–NH₂) in KH550 is more beneficial to the dispersion of ZnO nanoparticles in EG, which offers higher polarity than methanol.

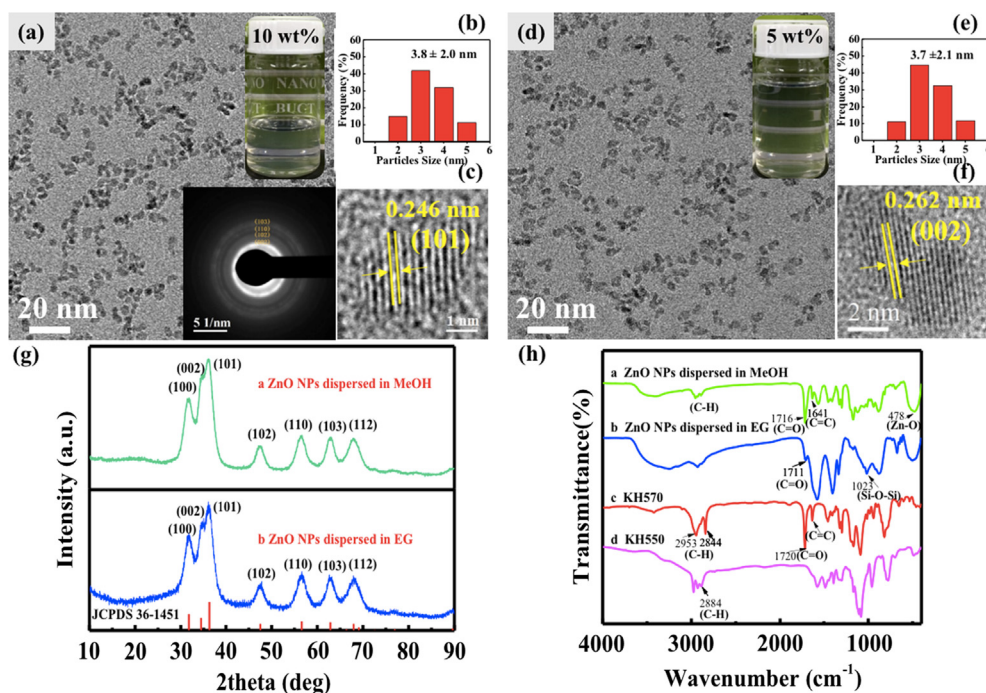
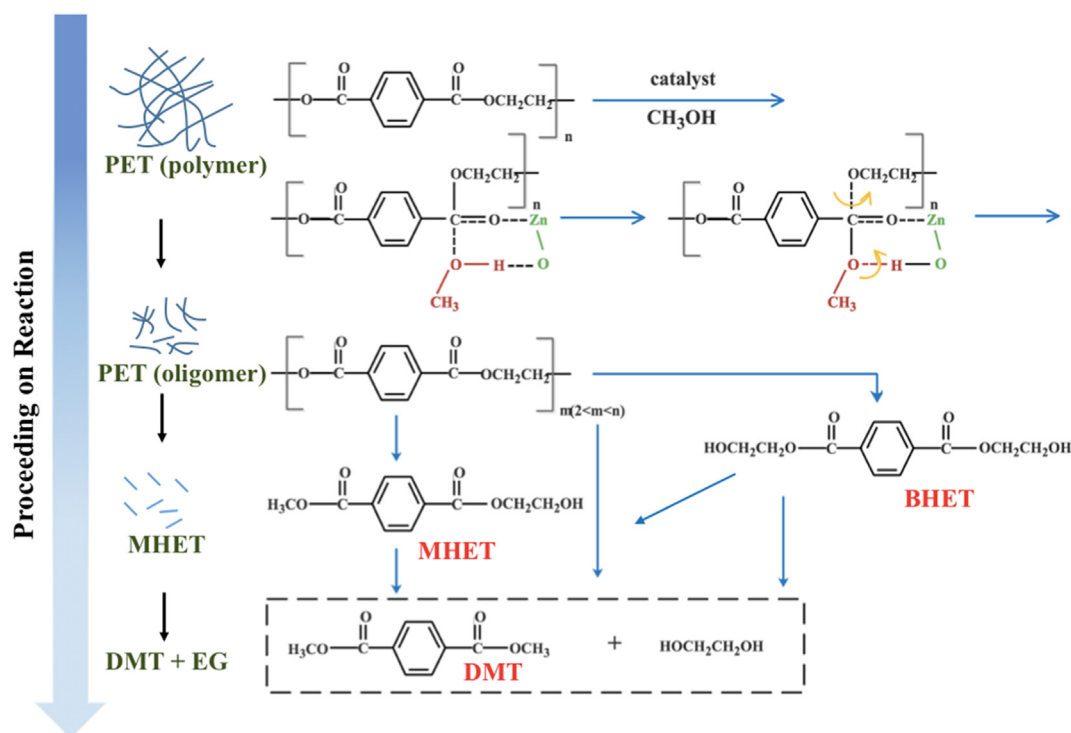


Fig. 1. TEM image (a) (the inset is a digital photograph of ZnO nanodispersions with a solid content of 10 wt% and the corresponding SAED pattern), particle size distributions (b) and HRTEM image (c) of methanol dispersion of ZnO nanoparticles. TEM image (d) (the inset is a digital photograph of ZnO nanodispersions with a solid content of 5 wt%), particle size distributions (e) and HRTEM image (f) of EG dispersion of ZnO nanoparticles. XRD patterns (g) of methanol and EG dispersions of ZnO nanoparticles. FTIR spectra (h) of methanol and EG dispersions of ZnO nanoparticles, KH570 and KH550.

In the following studies, the catalytic methanolysis activity of ZnO nanodispersions is explored in detail. According to the previous reports (López-Fonseca et al., 2010), it is known that PET is converted to oligomers, and the oligomers are then converted to bis-2-hydroxyethyl terephthalate (BHET) and 2-hydroxyethyl

methyl terephthalate (MHET) and, finally, to dimethyl terephthalate (DMT) and EG. Scheme 2 presents a possible catalytic mechanism of the methanolysis of PET. The oxygen of the carbonyl group (C=O) in the PET combines with the metal cations of the catalysts, thereby making the carbon cations become more electropositive.



Scheme 2. The methanolysis mechanism of PET over ZnO catalyst.

Then free lone pairs of electrons on the oxygen of methanol attack the carbon cation, resulting in the cleavage of the ester C-O bond (Saleh and Gupta, 2012; Saleh et al., 2016) and the formation of a new C-O bond with methanol. Moreover, the reaction products were dissolved in the mixture of methanol and trichloromethane and then analyzed by HPLC. Fig. S3 shows the clear separation of chromatogram peaks, thus elucidating the reasonability of HPLC analysis.

Fig. 2 shows the effects of reaction temperature in the range of 120–170 °C and time on the conversion of PET, the selectivity of BHET, MHET and the yield of DMT. As shown in Fig. 2a, the conversion of PET is only 2.5% at 120 °C (30 min), and then rapidly increases to 99.2% at 170 °C (30 min). This is because the methanolysis of PET with methanol is endothermic, leading to the significantly increased reaction rate with increasing the temperature. In Fig. 2b and c, the selectivity of MHET is much higher than that of BHET at the same reaction temperature and time. Moreover, the selectivity of BHET and MHET dramatically decreases with an increase in the reaction temperature. Simultaneously, the yield of DMT increases from 0% (120 °C, 30 min) to 95.0% (170 °C, 30 min) with increasing the temperature, as shown in Fig. 2d. More specifically, at a lower temperature of 120 °C, the yield of DMT is very low and increases slightly in 2 h. When the temperature is increased to 140 °C, the yield obviously increases, exceeding 50% at 2 h. When the temperature is further increased to 160 and 170 °C, the yield of DMT reaches 73% at 30 min and 92% at 15 min, respectively. The above results completely indicate that the formation of MHET is preferential to BHET in this catalytic process. And high temperature is beneficial to the yield of DMT. This might be because the transition step of BHET to DMT has a higher activation energy than the formation step of BHET in a multi-step series reaction (Genta et al., 2005).

Fig. 3 presents the effect of the concentration of catalyst on the conversion of PET, the selectivity of BHET, MHET, and the yield of DMT. Clearly, the conversion of PET increases linearly from 15.2% to 62.5% with the increased amount of ZnO catalyst from 0.1% to

1.3% (Fig. 3a). The selectivity of BHET and MHET has little change when the concentration of catalyst is lower than 0.7%. However, further increasing the catalyst concentration leads to the rapid decrease of the selectivity of BHET and MHET (Fig. 3b and c). Furthermore, the increase of catalyst concentration leads to an increase of yield of DMT from 5.1% to 52.4% (Fig. 3d). The above results indicate that this process can just be greatly accelerated when the catalyst concentration exceeds 0.7%. And the catalyst has a more obvious catalytic action on the transition step of BHET to DMT.

Fig. 4 depicts the effect of the weight ratio of methanol to PET on the conversion of PET, the selectivity of BHET, MHET, and the yield of DMT. It can be observed that the conversion of PET and the yield of DMT increase markedly with an increase in the weight ratio (Fig. 4a and d). Among them, the biggest increase is achieved when the weight ratio is increased from 5 to 6. Considering increasing the weight ratio of methanol to PET demands larger capital investment and energy consumption. Based on experimental data, the optimal weight ratio of methanol to PET is 6. Furthermore, the selectivity of BHET and MHET first slightly increases as the weight ratio of methanol to PET increases from 4 to 5, and the maximum appears at the weight ratio of 5. As the weight ratio further increases from 5 to 7, the selectivity of BHET and MHET decreases markedly. This is because the higher reactant concentration is beneficial to the proceeding of this whole reaction.

Fig. 5 displays the SEM and TEM images of the residual PET at different reaction times to reveal an evolution of the methanolysis of PET. SEM images indicate that the size of PET powder decreases during the process of methanolysis. The insert (in Fig. 5a) shows that the surface of the virgin PET is relatively smooth. However, it can be observed from TEM images that the porous structure is gradually formed and the surface of PET becomes significantly more microporous during the process. This result can also be verified from the etched surface in the insert of Fig. 5d. This is because that depolymerization process occurred from external to inside area and the methanolysis products are not deposited on the

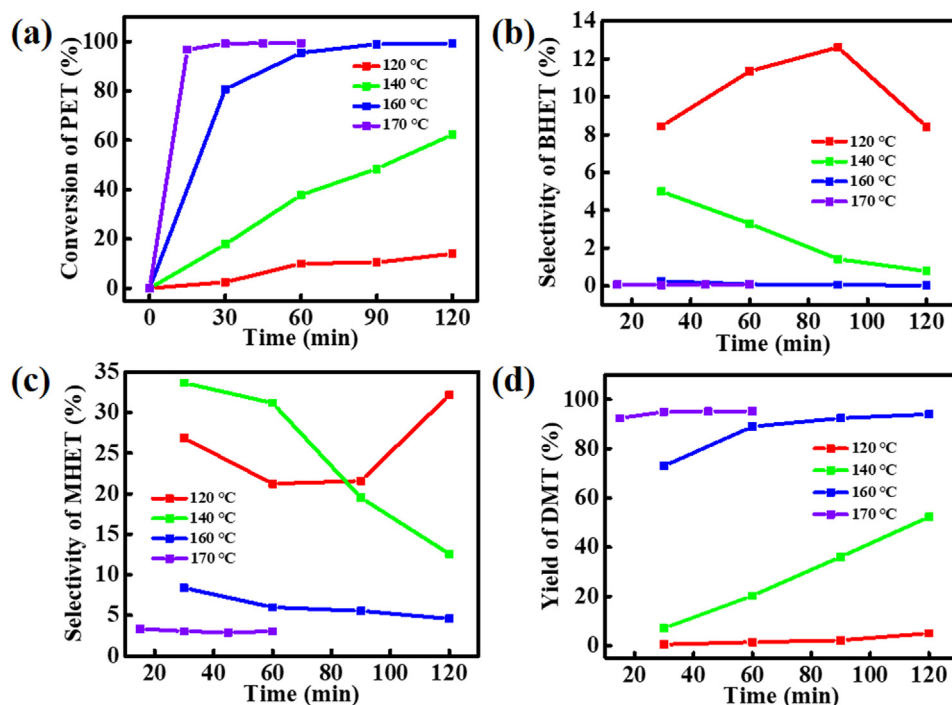


Fig. 2. The effects of reaction temperature and time on the conversion of PET (a) and the selectivity of BHET (b), MHET (c) and the yield of DMT (d). (the weight ratio of methanol to PET of 6, the weight ratio of catalyst to PET of 0.7 wt%).

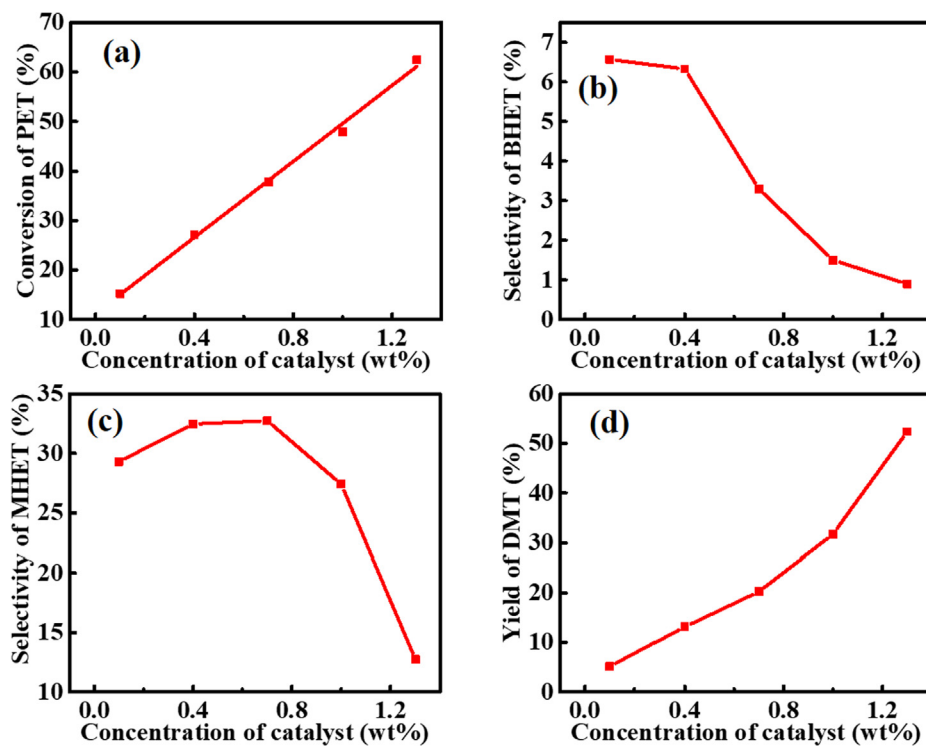


Fig. 3. The effect of the catalyst amount on the conversion of PET(a) and the selectivity of BHET (b), MHET (c) and the yield of DMT (d) (the weight ratio of methanol to PET of 6, 140 °C, 1 h).

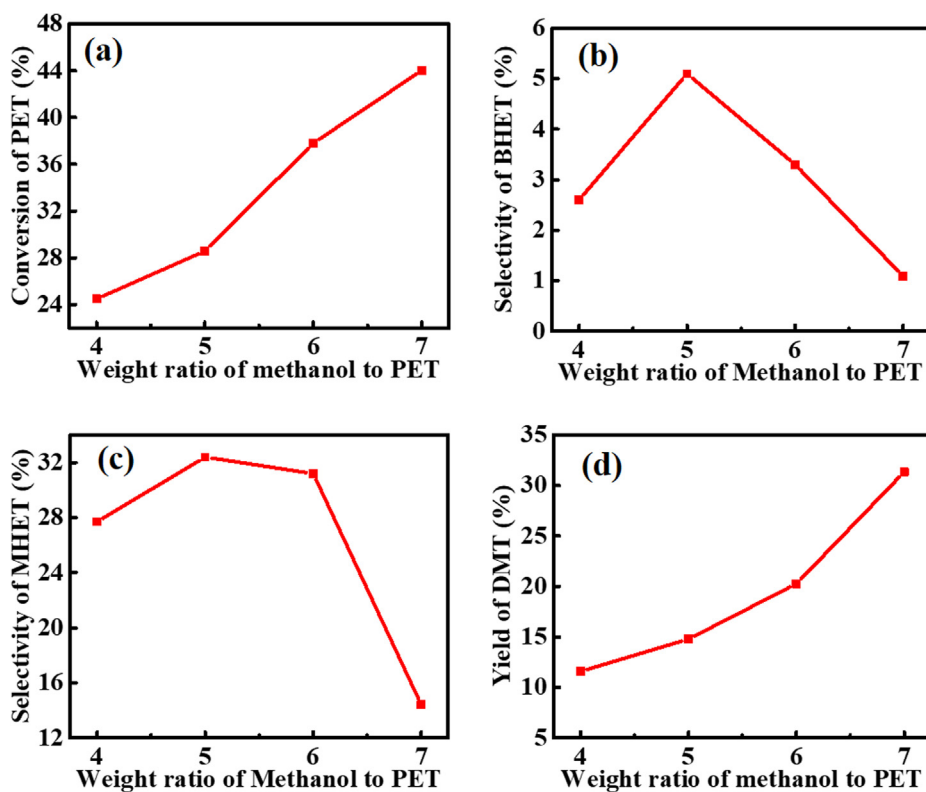


Fig. 4. The effect of the weight ratio of methanol to PET on the conversion of PET (a) and the selectivity of BHET (b), MHET (c) and the yield of DMT (d) (the weight ratio of catalyst to PET of 0.7 wt%, 140 °C, 1 h).

surface of PET. The formation of the porous framework not merely increased the effective reaction region but also facilitated penetra-

tion of methanol and catalyst into PET (Adio et al., 2017; Saleh, 2015a, 2015b, Saleh et al., 2017, 2018).

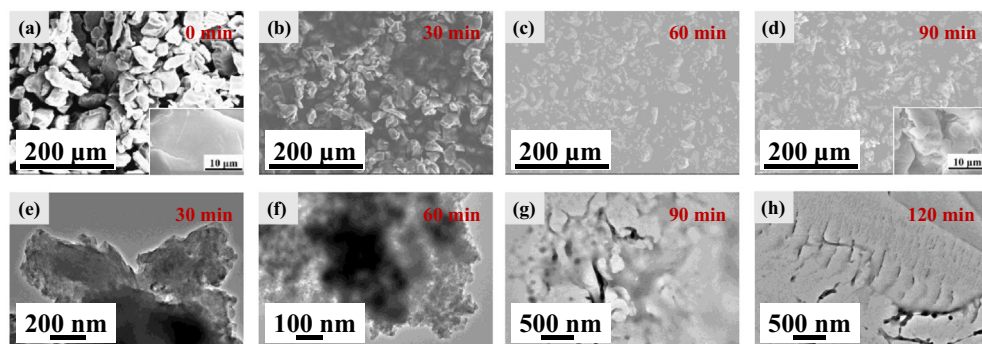


Fig. 5. SEM images of virgin PET (a) and residual PET after 30 min (b), 60 min (c) and 90 min (d) (the inset is a magnified photograph of the residual PET after 90 min). TEM images of the residual PET after 30 min (e), 60 min (f), 90 min (g) and 120 min (h) (the weight ratio of methanol to PET of 6, the weight ratio of catalyst to PET of 0.7 wt%, 140 °C).

Fig. 6 displays the FTIR spectra of raw PET, solid degradation products of PET materials and standard DMT to verify the functional group changes in the methanolysis process. The spectra show the C=O stretching at 1720 cm^{-1} , the ester C-O at 1252 cm^{-1} , -OH at 3432 cm^{-1} and alkyl C-H at 2844 and 2955 cm^{-1} (Saleh, 2016, 2018). The intense bands at 1435 and 1017 cm^{-1} are associated with the C=C stretching vibration of benzene and =CH bending vibration in-plane of benzene, respectively. The band at 1110 cm^{-1} is due to the C-O-C symmetrical stretching vibration. The FTIR spectra of residual PET and raw PET are nearly the same, which manifests that some structure of PET was unchanged during the depolymerization process (140 °C, 1 h). Combined with results in Fig. 5, it can be deduced that, at this stage, the internal structure of PET has not been destroyed in this process. In addition, the spectra of the mixture of DMT, MHET and BHET, DMT and standard DMT are similar because no new functional groups are formed in the depolymerization of PET. This result can also be verified from Scheme 2. However, as compared to the peak of the ester C-O at 1252 cm^{-1} of residual PET, the ester C-O stretching of DMT monomer shifted to 1283 cm^{-1} , mainly attributed to the COOCH_3 of DMT. This indicates that ester linkages are broken and replaced with COOCH_3 during methanolysis process.

Fig. 7 illustrates the TG, DTG and DSC curves of the residual PET obtained at different reaction times. A slight weight loss is observed between 120 and 200 °C, and the DTG curves show that the initial decomposition temperature (195, 189, 140, and

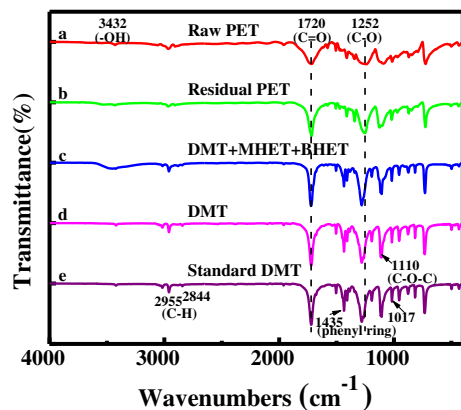


Fig. 6. FTIR spectra of raw PET (a), the residual PET (b) (the weight ratio of methanol to PET of 6, the weight ratio of catalyst to PET of 0.7 wt%, 140 °C, 1 h), the mixture of DMT, MHET and BHET (c) (the weight ratio of methanol to PET of 6, the weight ratio of catalyst to PET of 0.7 wt%, 140 °C, 1 h), the obtained DMT (d) (the weight ratio of methanol to PET of 6, the weight ratio of catalyst to PET of 0.7 wt%, 170 °C, 1 h) and standard DMT (e).

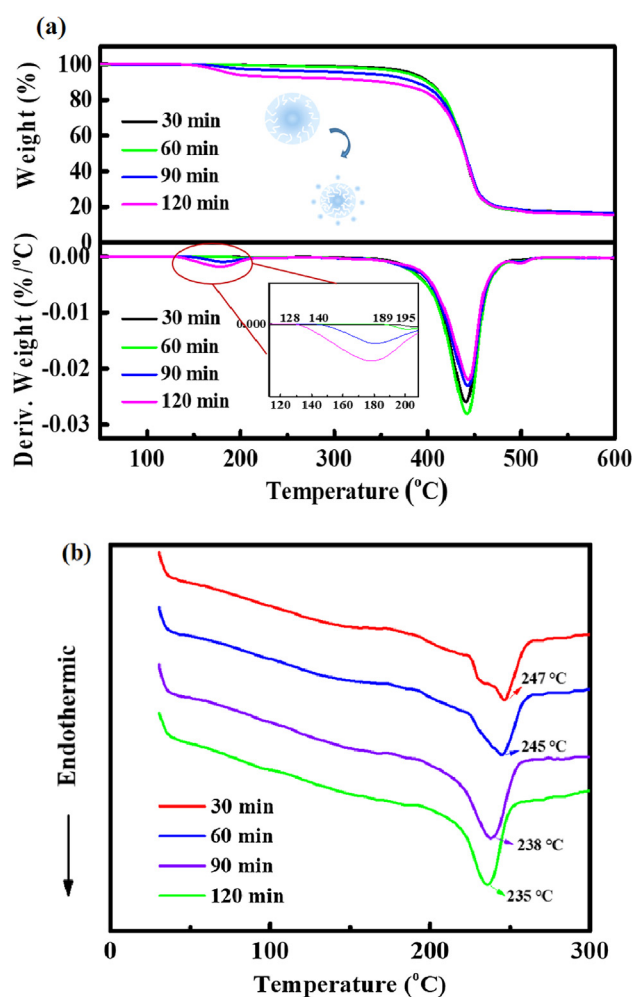


Fig. 7. TG, DTG (a) and DSC (b) curves of the residual PET after different times (the weight ratio of methanol to PET of 6, the weight ratio of catalyst to PET of 0.7 wt%, 140 °C).

128 °C) is shifted to lower temperature with the extension of the reactions. Furthermore, an obvious weight loss is observed between 250 and 450 °C, mainly ascribed to the random cleavage of ester linkages and the formation of various oligomers. The maximum decomposition temperature was almost unchanged. Combined with the SEM and TEM images in Fig. 5, it can be deduced that the internal structure of PET has not been destroyed in this

process. However, the surface of PET particles becomes more easily decomposed with the decreasing size and the formation of porous structure. In DSC curves, the endothermic peak at about 250 °C is related to the melting point of PET. In addition, the temperature corresponding to the endothermic peak also decreases obviously from 247 to 235 °C with the increase of reaction time from 30 to 120 min. From the results of TG and DSC analysis, it can be con-

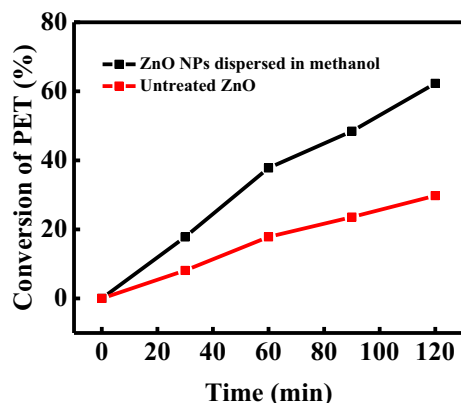


Fig. 8. Catalytic performance of methanol dispersion of ZnO nanoparticles and untreated ZnO nanoparticles (the weight ratio of methanol to PET of 6, the weight ratio of catalyst to PET of 0.7 wt%, 140 °C).

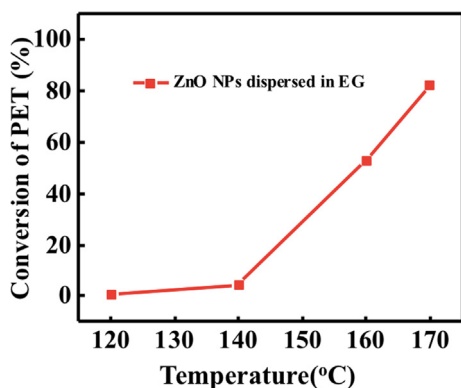


Fig. 9. The effect of reaction medium on the depolymerization reaction at different temperatures (the weight ratio of EG to PET of 6, the weight ratio of catalyst to PET of 0.7 wt%, 1 h).

cluded that PET is more easily decomposed since the structure of PET is gradually destroyed during the methanolysis process.

Fig. 8 compares the catalytic performance of methanol dispersion of ZnO nanoparticles and untreated ZnO nanoparticles. **Fig. S4** shows untreated ZnO nanoparticles have a severe agglomeration with an average size of about 20 nm. Compared with the ZnO nanoparticles dispersed in methanol, the width of XRD diffraction peak of untreated ZnO nanoparticles gets narrow with the absence of surface modifier, implying an increase in particle size, which is in agreement with above TEM results. The conversion of PET catalyzed by methanol dispersion of ZnO nanoparticles is about two times than that of untreated ZnO nanoparticles. This can mainly be attributed to the better dispersibility and stability of ZnO nanoparticles in reaction medium, as well as the more active sites from the ultrasmall size of ZnO nanoparticles. Furthermore, the pseudohomogeneous ZnO nanocatalysts could enter the porous framework, as shown in **Fig. 5g**, which is difficult for untreated ZnO nanoparticles with a larger size and severe agglomeration, to facilitate the depolymerization of PET.

Fig. 9 shows the glycolysis of PET over EG dispersion of ZnO nanoparticles under the same processing conditions. The conversion of PET depolymerized in EG is close to 0% at 120 °C and then increases to 82% at 170 °C in an hour with increasing temperature. As a comparison, the methanolysis rate of PET is much faster. This can be explained by the higher activity of methanol as a nucleophilic reagent. The steric hindrance provided by methanol as a nucleophile is smaller than that of EG, and the nucleophilic activity of the oxygen of alcoholic hydroxyl is higher. However, it is worth pointing out that glycolysis is more aligned with the green chemistry principles because of the lower toxicity of EG. The glycolysis and methanolysis of PET are very similar. Since they are essentially both transesterification reactions, so the influence of other reaction conditions and corresponding characterizations of glycolysis is not described in detail.

The ^1H and ^{13}C NMR spectra of obtained DMT (170 °C, 1 h) in **Fig. S5** indicate the high purity of DMT, which manifests the separability of pseudohomogeneous ZnO catalysts from products. **Fig. 10** shows the reusability of methanol dispersion of ZnO nanoparticles in the methanolysis of PET and TEM image of ZnO nanoparticles after five cycles. It can be seen that the conversion of PET decreases within five experiments because of the inevitable loss of catalysts in the process of separation and transfer. So it is desired to develop pseudohomogeneous nanocatalysts based on magnetic separation. Furthermore, the increasing concentration of EG in methanol solution might also have an unfavorable effect on the degradation of PET. The TEM image of ZnO nanoparticles shows that the ZnO still keeps monodispersed with nearly unchanged size after five catalytic cycles.

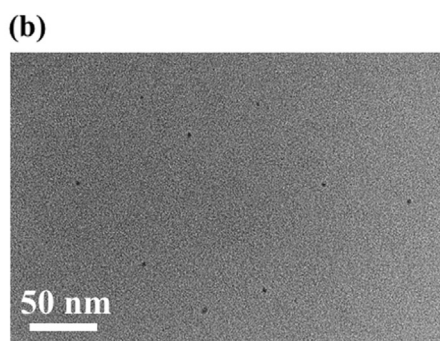
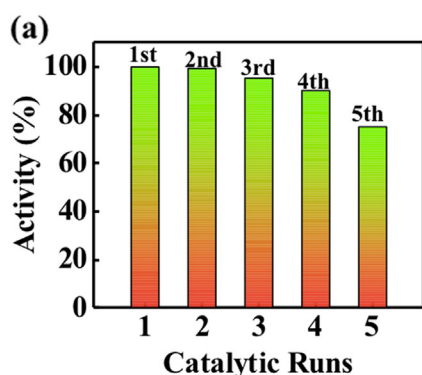


Fig. 10. The reusability of methanol dispersion of ZnO nanoparticles in the methanolysis of PET and TEM image of ZnO nanoparticles after five cycles.

Table 1

Comparison of reported catalysts for the chemical depolymerization of PET.

m _{PET} /m _{catalyst}	X/%	Reagent	Temperature/°C	Time/h	Catalyst	Activity ^d /h ⁻¹	Reference
143	80.7 ^a	MeOH	160	0.50	ZnO nanodispersions	231	This work
143	96.7 ^a	MeOH	170	0.25	ZnO nanodispersions	553	This work
143	82.3 ^a	EG	170	1.00	ZnO nanodispersions	117	This work
–	99.9 ^a	MeOH	300	0.50	–	–	(Genta et al., 2005)
–	96.6 ^b	MeOH	320	1.13	–	–	(Liu et al., 2015)
–	98.7 ^a	MeOH	270	0.67	–	–	(Yang et al., 2002)
100	81.3 ^c	EG	196	0.83	MgO–Al ₂ O ₃	97.6	(Chen et al., 2012)
100	92.0 ^c	EG	260	1.00	ZnMn ₂ O ₄	92.0	(Imran et al., 2013)
100	82.0 ^c	EG	196	0.33	ZnO–Al ₂ O ₃	246	(Chen et al., 2015)
333	100 ^a	EG	180	3.08	SO ₄ ²⁻ –ZnO–TiO ₂ –300°C	108	(Zhu et al., 2012)
20.0	95.0 ^c	EG	300	1.00	γ-Fe ₂ O ₃	19.0	(Bartolome et al., 2014)
20.0	100 ^c	EG	190	2.00	Fe ₃ O ₄ –MWCNT	10.0	(Al-Sabagh et al., 2016b)

^a Conversion of PET.^b Yield of DMT.^c Yield of BHET.^d Calculated by employing equation: Activity = m_{PET}•X•m_{catalyst}⁻¹•Time⁻¹.

Table 1 compares some reported catalysts for the chemical depolymerization of PET with this work. The activity was defined as the mass of converted PET or received BHET per unit mass of active components of catalyst per hour for better comparison of different reaction systems. It was determined that the catalytic activity was 553 and 117 g PET h⁻¹ (g ZnO)⁻¹ for depolymerization of PET occurred in methanol and EG, respectively. And the former had a higher conversion and a shorter reaction time than the latter. Furthermore, compared with supercritical conditions (T > 270 °C) (Genta et al., 2005; Liu et al., 2015; Yang et al., 2002), the reaction temperature and time were obviously reduced by adopting pseudohomogeneous ZnO nanocatalyst. For the depolymerization process in EG, compared with other oxide catalysts (Al-Sabagh et al., 2016b; Bartolome et al., 2014; Chen et al., 2012, 2015; Imran et al., 2013; Zhu et al., 2012), the pseudohomogeneous ZnO catalyst exhibited a higher activity at a lower temperature (170 °C) except ZnO–Al₂O₃. Such an excellent depolymerization performance should be ascribed to the high stability and excellent dispersity of ultrasmall ZnO nanoparticles in the reaction reagent, which provide a complete contact between the reactant and the catalyst with little diffusion resistance.

4. Conclusions

In summary, highly stable methanol and EG nanodispersion of ultrasmall ZnO nanoparticles with a size of about 4 nm were successfully synthesized and firstly adopted as pseudohomogeneous nanocatalysts for the catalytic alcoholysis of PET. The as-prepared monodispersed ZnO nanoparticles have high catalytic activity for the methanolysis and glycolysis of PET. In the methanolysis process, higher temperature (>160 °C) benefited the conversion of PET and the yield of DMT. The conversion of PET reached up to 97% and the yield of DMT was up to 95% at 170 °C at 15 min. The activities of 553 g PET h⁻¹ (g ZnO)⁻¹ can be achieved. The higher catalyst/PET and methanol/PET weight ratio could accelerate the reaction, thereby increasing the yield of DMT monomer. Degradation reaction first occurs at external area of PET and ultrasmall ZnO nanocatalyst can enter the microporous structure to speed up the reaction. Furthermore, the catalytic methanolysis of PET have a shorter reaction time (1/4), a higher conversion and activity (about 4.7 times) than the glycolysis process at the same reaction temperature (170 °C). The above results completely indicate that the pseudohomogeneous ZnO nanocatalysts are efficient, easily recoverable and reusable, making the process more sustainable and environment-friendly. Therefore, it could be envisioned that this pseudohomogeneous nanocatalyst may be a promising strategy for the chemical depolymerization of PET.

CRediT authorship contribution statement

Jin-Tao Du: Investigation, Writing - original draft. **Qian Sun:** Writing - review & editing. **Xiao-Fei Zeng:** Formal analysis. **Dan Wang:** Methodology. **Jie-Xin Wang:** Supervision, Writing - review & editing, Funding acquisition. **Jian-Feng Chen:** Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was financially supported by National Key R & D Program of China (2017YFA0206801) and National Natural Science Foundation of China (21622601 and 21878015). Qian Sun is grateful to the International Postdoctoral Exchange Fellowship Program (Talent-Introduction Program) for funding, jointly provided by the Office of China Postdoc Council and Beijing University of Chemical Technology.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ces.2020.115642>.

References

- Adio, S.O., Omar, M.H., Asif, M., Saleh, T.A., 2017. Arsenic and selenium removal from water using biosynthesized nanoscale zero-valent iron: a factorial design analysis. *Process Saf. Environ.* 107, 518–527.
- Ali, S.A., Rachman, I.B., Saleh, T.A., 2017. Simultaneous trapping of Cr(III) and organic dyes by a pH-responsive resin containing zwitterionic aminomethyl phosphonate ligands and hydrophobic pendants. *Chem. Eng. J.* 330, 663–674.
- Al-Sabagh, A.M., Yehia, F.Z., Eshaq, G., ElMetwally, A.E., 2015. Ionic liquid-coordinated ferrous acetate complex immobilized on bentonite as a novel separable catalyst for PET glycolysis. *Ind. Eng. Chem. Res.* 54, 12474–12481.
- Al-Sabagh, A.M., Yehia, F.Z., Eshaq, G., Rabie, A.M., ElMetwally, A.E., 2016a. Greener routes for recycling of polyethylene terephthalate. *Egypt. J. Pet.* 25, 53–64.
- Al-Sabagh, A.M., Yehia, F.Z., Harding, D.R.K., Eshaq, G., ElMetwally, A.E., 2016b. Fe₃O₄-boosted MWCNT as an efficient sustainable catalyst for PET glycolysis. *Green Chem.* 18, 3997–4003.
- Awaja, F., Pavel, D., 2005. Recycling of PET. *Eur. Polym. J.* 41, 1453–1477.
- Bartolome, L., Imran, M., Lee, K.G., Sangalang, A., Ahn, J.K., Kim, D.H., 2014. Superparamagnetic γ-Fe₂O₃ nanoparticles as an easily recoverable catalyst for the chemical recycling of PET. *Green Chem.* 16, 279–286.
- Chen, F., Wang, G., Li, W., Yang, F., 2012. Glycolysis of poly(ethylene terephthalate) over Mg–Al mixed oxides catalysts derived from hydrotalcites. *Ind. Eng. Chem. Res.* 52, 565–571.

- Chen, F., Zhou, Q., Bu, R., Yang, F., Li, W., 2015. Kinetics of poly(ethylene terephthalate) fiber glycolysis in ethylene glycol. *Fibers and Polym.* 16, 1213–1219.
- Du, S., Valla, J.A., Parnas, R.S., Bollas, G.M., 2016. Conversion of polyethylene terephthalate based waste carpet to benzene-rich oils through thermal, catalytic, and catalytic steam pyrolysis. *ACS Sustainable Chem. Eng.* 4, 2852–2860.
- Genta, M., Goto, M., Sasaki, M., 2010. Heterogeneous continuous kinetics modeling of PET depolymerization in supercritical methanol. *J. Supercrit. Fluids* 52, 266–275.
- Genta, M., Iwaya, T., Sasaki, M., Goto, M., Hirose, T., 2005. Depolymerization mechanism of poly(ethylene terephthalate) in supercritical methanol. *Ind. Eng. Chem. Res.* 44, 3894–3900.
- George, N., Kurian, T., 2014. Recent developments in the chemical recycling of postconsumer poly(ethylene terephthalate) waste. *Ind. Eng. Chem. Res.* 53, 14185–14198.
- Geyer, B., Lorenz, G., Kandelbauer, A., 2016. Recycling of poly(ethylene terephthalate) – a review focusing on chemical methods. *Express Polym. Lett.* 10, 559–586.
- Goto, M., Koyamoto, H., Kodama, A., Hirose, T., Nagaoka, S., McCoy, B.J., 2002. Degradation kinetics of polyethylene terephthalate in supercritical methanol. *AIChE J.* 48, 136–144.
- Hoang, C.N., Le, T.T.N., Hoang, Q.D., 2018. Glycolysis of poly(ethylene terephthalate) waste with diethyleneglycol under microwave irradiation and $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$ catalyst. *Polym. Bull.* 76, 23–34.
- Iannone, F., Casiello, M., Monopoli, A., Cotugno, P., Sportelli, M.C., Picca, R.A., Cioffi, N., Dell'Anna, M.M., Nacci, A., 2017. Ionic liquids/ ZnO nanoparticles as recyclable catalyst for polycarbonate depolymerization. *J. Mol. Catal. A: Chem.* 426, 107–116.
- Imran, M., Kim, D.H., Al-Masry, W.A., Mahmood, A., Hassan, A., Haider, S., Ramay, S. M., 2013. Manganese-, cobalt-, and zinc-based mixed-oxide spinels as novel catalysts for the chemical recycling of poly(ethylene terephthalate) via glycolysis. *Polym. Degrad. Stab.* 98, 904–915.
- Joo, S., Cho, I.J., Seo, H., Son, H.F., Sagong, H.Y., Shin, T.J., Choi, S.Y., Lee, S.Y., Kim, K.J., 2018. Structural insight into molecular mechanism of poly(ethylene terephthalate) degradation. *Nat. Commun.* 9, 382.
- Karayannidis, G.P., Achilias, D.S., 2007. Chemical Recycling of Poly(ethylene terephthalate). *Macromol. Mater. Eng.* 292, 128–146.
- Kurokawa, H., Ohshima, M.-A., Sugiyama, K., Miura, H., 2003. Methanolysis of polyethylene terephthalate (PET) in the presence of aluminium triisopropoxide catalyst to form dimethyl terephthalate and ethylene glycol. *Polym. Degrad. Stab.* 79, 529–533.
- Lin, B., Lin, Z., Chen, S., Yu, M., Li, W., Gao, Q., Dong, M., Shao, Q., Wu, S., Ding, T., Guo, Z., 2019a. Surface intercalated spherical $\text{MoS}_2\text{Se}_{2(1-x)}$ nanocatalysts for highly efficient and durable hydrogen evolution reactions. *Dalton Trans.* 48, 8279–8287.
- Lin, Z., Lin, B., Wang, Z., Chen, S., Wang, C., Dong, M., Gao, Q., Shao, Q., Ding, T., Liu, H., Wu, S., Guo, Z., 2019b. Facile preparation of 1T/2H- $\text{Mo}(\text{S}_{1-x}\text{Se}_x)_2$ nanoparticles for boosting hydrogen evolution reaction. *ChemCatChem* 11, 2217–2222.
- Liu, Q., Li, R., Fang, T., 2015. Investigating and modeling PET methanolysis under supercritical conditions by response surface methodology approach. *Chem. Eng. J.* 270, 535–541.
- López-Fonseca, R., Duque-Ingunza, I., de Rivas, B., Arnaiz, S., Gutiérrez-Ortiz, J.L., 2010. Chemical recycling of post-consumer PET wastes by glycolysis in the presence of metal salts. *Polym. Degrad. Stab.* 95, 1022–1028.
- Lorenzetti, C., Manaresi, P., Berti, C., Barbiroli, G., 2006. Chemical recovery of useful chemicals from polyester (PET) waste for resource conservation: a survey of state of the art. *J. Polym. Environ.* 14, 89–101.
- Malik, N., Kumar, P., Shrivastava, S., Ghosh, S.B., 2016. An overview on PET waste recycling for application in packaging. *Int. J. Plast. Technol.* 21, 1–24.
- Mishra, S., Goje, A.S., 2003. Kinetic and thermodynamic study of methanolysis of poly(ethylene terephthalate) waste powder. *Polym. Int.* 52, 337–342.
- Monsigny, L., Berthet, J.-C., Cantat, T., 2018. Depolymerization of waste plastics to monomers and chemicals using a hydrosilylation strategy facilitated by brookhart's iridium(III) catalyst. *ACS Sustainable Chem. Eng.* 6, 10481–10488.
- Paszun, D., Spychar, T., 1997. Chemical recycling of poly (ethylene terephthalate). *Ind. Eng. Chem. Res.* 36, 1373–1383.
- Polk, M.B., Leboeuf, L.L., Shah, M., Won, C.-Y., Hu, X., Ding, W., 1999. Nylon 66, Nylon 46, and pet phase-transfer-catalyzed alkaline depolymerization at atmospheric pressure. *Polym. Plast. Technol.* 38, 459–470.
- Saleh, T.A., 2015a. Isotherm, kinetic, and thermodynamic studies on Hg(II) adsorption from aqueous solution by silica- multiwall carbon nanotubes. *Environ. Sci. Pollut. R. I* 22, 16721–16731.
- Saleh, T.A., 2015b. Mercury sorption by silica/carbon nanotubes and silica/activated carbon: a comparison study. *J. Water Supply Res. T.* 64, 892–903.
- Saleh, T.A., 2016. Nanocomposite of carbon nanotubes/silica nanoparticles and their use for adsorption of Pb(II) : from surface properties to sorption mechanism. *Desalin. Water Treat.* 57, 10730–10744.
- Saleh, T.A., 2018. Simultaneous adsorptive desulfurization of diesel fuel over bimetallic nanoparticles loaded on activated carbon. *J. Clean. Prod.* 172, 2123–2132.
- Saleh, T.A., Gupta, V.K., 2012. Synthesis and characterization of alumina nanoparticles polyamide membrane with enhanced flux rejection performance. *Sep. Purif. Technol.* 89, 245–251.
- Saleh, T.A., Muhammad, A.M., Ali, S.A., 2016. Synthesis of hydrophobic cross-linked polyzwitterionic acid for simultaneous sorption of Eriochrome black T and chromium ions from binary hazardous waters. *J. Colloid Interf. Sci.* 468, 324–333.
- Saleh, T.A., Naeemullah, Tuzen, M., Sari, A., 2017. Polyethylenimine modified activated carbon as novel magnetic adsorbent for the removal of uranium from aqueous solution. *Chem. Eng. Res. Des.* 117, 218–227.
- Saleh, T.A., Tuzen, M., Sari, A., 2018. Polyamide magnetic polygorskite for the simultaneous removal of Hg(II) and methyl mercury; with factorial design analysis. *J. Environ. Manage.* 211, 323–333.
- Veregue, F.R., Pereira da Silva, C.T., Moisés, M.P., Meneguín, J.G., Guilherme, M.R., Arroyo, P.A., Favaro, S.L., Radovanovic, E., Giroto, E.M., Rinaldi, A.W., 2018. Ultrasmall cobalt nanoparticles as a catalyst for PET glycolysis: a green protocol for pure hydroxyethyl terephthalate precipitation without water. *ACS Sustainable Chem. Eng.* 6, 12017–12024.
- Wan, B.Z., Kao, C.Y., Cheng, W.-H., 2001. Kinetics of depolymerization of poly (ethylene terephthalate) in a potassium hydroxide solution. *Ind. Eng. Chem. Res.* 40, 509–514.
- Wang, Q., Geng, Y., Lu, X., Zhang, S., 2015. First-row transition metal-containing ionic liquids as highly active catalysts for the glycolysis of poly(ethylene terephthalate) (PET). *ACS Sustainable Chem. Eng.* 3, 340–348.
- Wang, Y., Zhang, Y., Song, H., Wang, Y., Deng, T., Hou, X., 2019. Zinc-catalyzed ester bond cleavage: chemical degradation of polyethylene terephthalate. *J. Clean. Prod.* 208, 1469–1475.
- Yamaye, M., Hashime, T., Yamamoto, K., Kosugi, Y., Cho, N., Ichiki, T., Kito, T.J.L., Research, E.C., 2002. Chemical recycling of poly(ethylene terephthalate). 2. Preparation of terephthalohydroxamic acid and terephthalohydrazide. *Ind. Eng. Chem. Res.* 41, 3993–3998.
- Yang, Y., Lu, Y., Xiang, H., Xu, Y., Li, Y.J.P.D., 2002. Study on methanolytic depolymerization of PET with supercritical methanol for chemical recycling. *Polym. Degrad. Stab.* 75, 185–191.
- Zhang, X., Fevre, M., Jones, G.O., Waymouth, R.M., 2018. Catalysis as an Enabling Science for Sustainable Polymers. *Chem. Rev.* 118, 839–885.
- Zhu, M., Li, S., Li, Z., Lu, X., Zhang, S., 2012. Investigation of solid catalysts for glycolysis of polyethylene terephthalate. *Chem. Eng. J.* 185–186, 168–177.